Richwell Cyrille Santos Perez

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EDUCATION

University of Illinois Urbana-Champaign August 2022 - May 2023
Master of Computer Science GPA: 3.52
University of Illinois Urbana-Champaign August 2018 - May 2023
Bachelor of Science in Computer Science with Honors GPA: 3.81

EXPERIENCE

Safran Laramie, WY; Brea, CA
AI Engineer June 2025 - Present

- Designed and deployed a full-stack RAG chatbot with Vue.js, Quart, and PostgreSQL, integrated via Azure App Services to serve 600+ internal users with scalable, reliable AI-driven responses
- Reduced response latency by 50% (2× faster) through optimization of multi-agent orchestration, caching, and prompt distillation
- Deployed scalable RESTful APIs with automated testing, structured logging, and performance monitoring to enhance reliability
- Applied ML methods (K-Means, DBSCAN, topic modeling) to cluster RAG conversation data and built a monitoring dashboard to visualize insights and continuously improve
- Building an ML-based predictive maintenance system to forecast RDU (Removable Display Unit) resets in in-flight entertainment (IFE) systems, using historical telemetry and operational data to improve reliability and minimize downtime
- Developing a Copilot agent in Microsoft Copilot Studio, integrating enterprise data and productivity platforms to enable unified access to analytics, collaboration, and ERP systems through AI-driven automation

Illinois Secretary of State Springfield, IL
Database Administrator February 2025 - June 2025

- Managed high-throughput DB2 databases (5 B+ records) on z/OS mainframe systems, optimizing queries and schemas to support statewide applications
- Improved query performance by designing and tuning complex SQL queries and schemas, reducing lookup time from 20 seconds to instantaneous for critical backend application queries
- Implemented ETL pipelines and automation scripts to streamline data migration and significantly reduce manual effort
- Collaborated with cross-functional development teams to design efficient schemas, stored procedures, and to integrate Azure-based services for scalable, cloud-connected applications
- Supported the development and deployment of statewide digital initiatives, including mobile driver's licenses and REAL ID

University of Illinois Urbana-Champaign Urbana, IL
Graduate Teaching Assistant August 2022 - May 2023

- Led discussions and labs for Software Design & Database Systems, teaching object-oriented design, design patterns, and database performance optimization to 800+ students

Software Engineer July 2021 - August 2021

- Developed LabWindows/CVI software to support new magnet mapping hardware, implementing real-time data interpretation, control logic, and optimizing frontend/backend performance

PROJECTS

UIUC Letter Grades — Full-Stack Web App with AI Integration

- Built a full-stack analytics platform with Node.js, MongoDB, and Python to visualize and analyze GPA data across 10+ years, enabling students to compare instructors and predict course outcomes
- Trained ML models (linear/logistic regression, sequential models) and visualizations for predictive grade analytics
- Analyzed and designed interactive data visualizations for users to track performance across semesters and predict future grades
- Implemented user authentication and automated data processing pipelines for efficient data handling

Video Anomaly Detection Model — Leveraging Weapon Recognition for Enhanced Accuracy

- Integrated weapon detection with anomaly scores, improving crime classification accuracy through multi-model decision-making
- Developed an AI pipeline using UCF-Crime anomaly detection & YOLO/Faster R-CNN for real-time crime recognition
- Optimized anomaly detection using Multi-Instance Learning (MIL) & improved loss functions for better predictions

Deep Learning Aerosol Model — *3rd Place, Ashby Prize in Computational Science, NCSA AI Innovation Hackathon*

- Engineered a transformer and Encoder/Decoder models to capture temporal patterns in climate-relevant aerosol properties
- Applied advanced data preprocessing techniques to handle high-dimensional climate data and reduce the curse of dimensionality
- Leveraged HAL 9000 (UIUC's supercomputer) for high-performance training, optimizing deep learning workload scalability

SKILLS

Languages and Software: Python, Java, C/C++, JavaScript, Typescript, SQL, DB2, MongoDB, PostgreSQL, Neo4j, Bash/Shell
AI and Data Science: TensorFlow, PyTorch, Keras, Scikit-learn, Langchain, Pandas, NumPy, Matplotlib
Frameworks and Tools: GCP, Azure, AWS, React.js, Node.js, Git, CI/CD, Docker, Visual Studio, Postman, Power BI